



# OneEDGE Connection Agent User Guide

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## APPLICABILITY TABLE

### PRODUCTS

- ■ ME910C1-E2
- ■ ME910C1-NA
- ■ ME910C1-E1
- ■ ME910C1-AU
- ■ ME910C1-J1
- ■ ME910C1-K1
- ■ ME910C1-WW
- ■ ME910C1-P1
- ■ ME910C1-WW
- ■ LE910C1-EUX
  
- ■ ME310G1-W1
- ■ ME310G1-WW
  
- ■ ME910G1-W1
- ■ ME910G1-WW
  
- ■ ML865G1-WW
- ■ ME310G1-W2

## CONTENTS

|                                                                |           |
|----------------------------------------------------------------|-----------|
| <b>NOTICES LIST .....</b>                                      | <b>2</b>  |
| <b>COPYRIGHTS .....</b>                                        | <b>2</b>  |
| <b>COMPUTER SOFTWARE COPYRIGHTS .....</b>                      | <b>2</b>  |
| <b>USAGE AND DISCLOSURE RESTRICTIONS .....</b>                 | <b>3</b>  |
| I. License Agreements .....                                    | 3         |
| II. Copyrighted Materials .....                                | 3         |
| III. High Risk Materials .....                                 | 3         |
| IV. Trademarks .....                                           | 3         |
| V. Third Party Rights .....                                    | 3         |
| <b>APPLICABILITY TABLE .....</b>                               | <b>4</b>  |
| <b>CONTENTS .....</b>                                          | <b>5</b>  |
| <b>1. INTRODUCTION .....</b>                                   | <b>7</b>  |
| 1.1. Scope .....                                               | 7         |
| 1.2. Audience.....                                             | 7         |
| 1.3. Contact Info and Support .....                            | 7         |
| 1.4. Icons and Text Conventions .....                          | 8         |
| 1.5. Related Documents .....                                   | 9         |
| <b>2. OVERVIEW .....</b>                                       | <b>10</b> |
| 2.1. Introduction.....                                         | 10        |
| 2.2. Hostless Approach.....                                    | 10        |
| 2.3. Hosted .....                                              | 11        |
| 2.4. Limitations .....                                         | 12        |
| <b>3. AT COMMANDS REFERENCES .....</b>                         | <b>13</b> |
| 3.1. AT#CASN - Start and Stop the Connection Agent .....       | 16        |
| 3.2. AT#CAVER - Check the version of the connection agent..... | 17        |
| 3.3. AT#CACFG - Connection Agent parameters .....              | 17        |
| 3.4. AT#CASAVE - Save Parameter of the Connection Agent.....   | 26        |
| 3.5. AT#CASTATS - Connection Agent Statistics .....            | 26        |
| <b>4. URC MESSAGES.....</b>                                    | <b>28</b> |
| 4.1. Preliminary info .....                                    | 28        |
| 4.2. 3GPP events .....                                         | 29        |
| 4.3. Antenna Signal .....                                      | 32        |
| 4.4. Network Registration .....                                | 32        |

|           |                                                   |           |
|-----------|---------------------------------------------------|-----------|
| 4.5.      | PDP context.....                                  | 32        |
| 4.6.      | System Codes .....                                | 32        |
| <b>5.</b> | <b>CONFIGURABLE LWM2M PARAMETERS.....</b>         | <b>33</b> |
| <b>6.</b> | <b>USE CASES.....</b>                             | <b>35</b> |
| 6.1.      | Preliminary – Load Application.....               | 35        |
| 6.2.      | Use case #1 – PDP Always connected .....          | 38        |
| 6.3.      | Use case #2 – Always registered to a network..... | 39        |
| <b>7.</b> | <b>ACRONYMS.....</b>                              | <b>41</b> |
| <b>8.</b> | <b>DOCUMENT HISTORY .....</b>                     | <b>44</b> |

# 1. INTRODUCTION

## 1.1. Scope

This document covers the Telit's Connection Agent features and usage.

## 1.2. Audience

The present User Guide is addressed to users that need to use Connection Agent AT interface and its functionalities. The reader can learn the use of the AT commands through simple examples shown in the document.

## 1.3. Contact Info and Support

For general contact, technical support services, technical questions and report documentation errors contact Telit Technical Support at:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com (for Short Range Devices)

Alternatively, use:

<http://www.telit.com/support>

For detailed information about where you can buy the Telit modules or for recommendations on accessories and components visit:

<http://www.telit.com>

Our aim is to make this guide as helpful as possible. Keep us informed of your comments and suggestions for improvements.

Telit appreciates feedback from the users of our information.

## 1.4. Icons and Text Conventions



SET section – This section provides all information related to SET functionality of involved AT command. If it has got strictly and relevant SET information, these are located at section end.



READ section – This section provides all information related to READ functionality of involved AT command. If it has got strictly and relevant READ information, these are located at section end.



TEST section – This section provides all information related to TEST functionality of involved AT command. If it has got strictly and relevant TEST information, these are located at section end.



Additional info – This section provides any kind of additional and useful information related to the AT command section as well as command exceptions or special behavior cases.



REFERENCE section – This section provides useful references (standards or normative) related to involved AT command.



EXAMPLE section – This section provides useful examples related to involved AT command.



NOTE section – This section provides all information related to involved AT commands. Each note can provide a different level of information: danger, caution/warning and tip/information.



Danger – This information **MUST** be followed or catastrophic equipment failure or bodily injury may occur.



Caution or Warning – Alerts the user to important points about integrating the module, if these points are not followed, the module and end user equipment may fail or malfunction.



Tip or Information – Provides advice and suggestions that may be useful when integrating the module.

All dates are in ISO 8601 format, i.e. YYYY-MM-DD.



## 1.5. Related Documents

- [1] ME910C1 AT Command Reference Guide: 80529ST10815A
- [2] 3GPP TS 24.008 V10.10.0 (2013-03)
- [3] GSMA CLP.03 IoT Device Connection Efficiency Guidelines Ver. 2.0 01 July 2015
- [4] ME910C1 HW User Guide: 1VV0301351
- [5] ME910C1 LwM2M AT commands User Guide: 80529ST10974A
- [6] OMA Lightweight M2M Specification

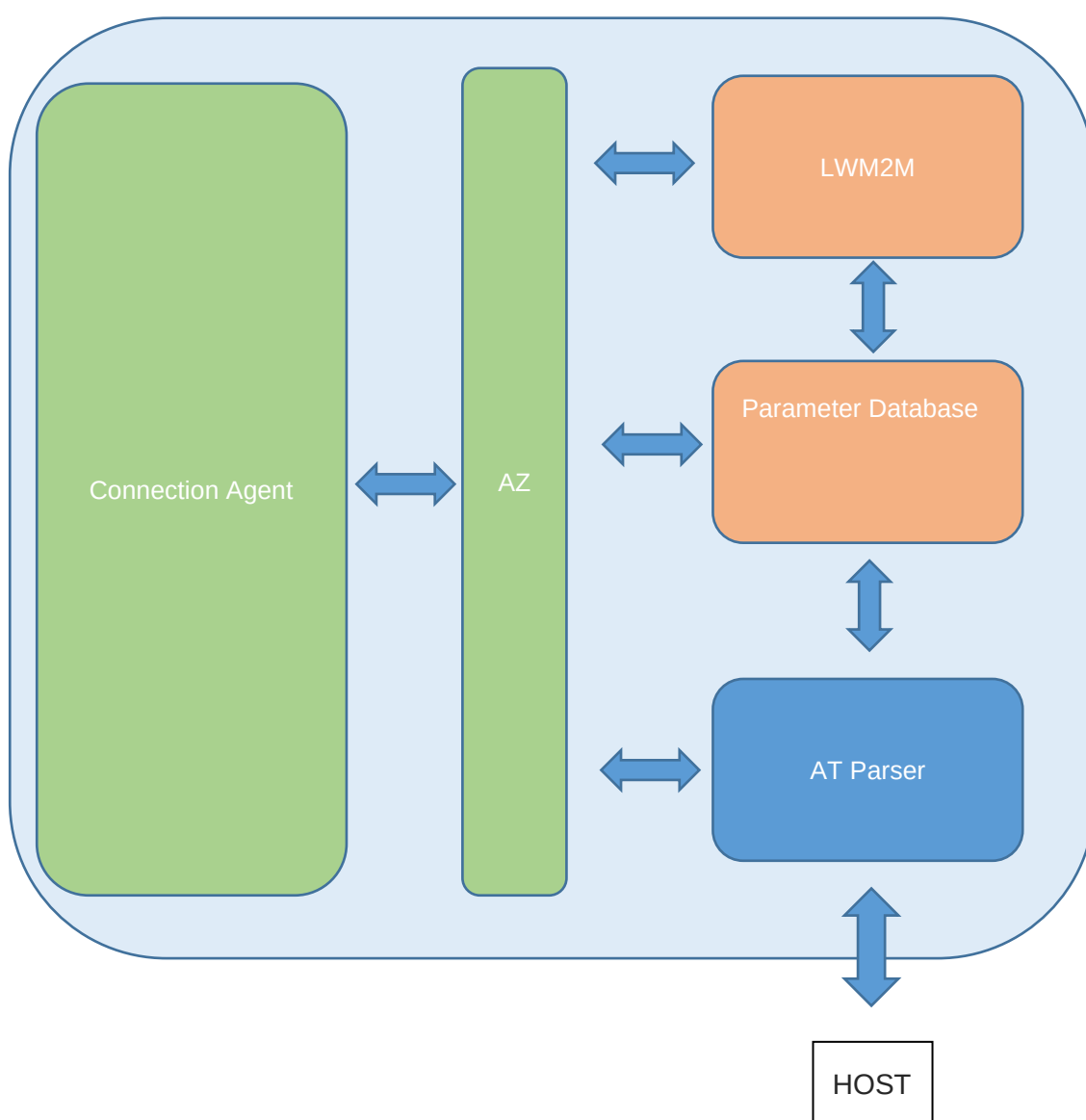
## 2. OVERVIEW

### 2.1. Introduction

Connection Agent is a software application that runs on the module, and allows the user to automatically configure and enable the data connection, keeping it enabled whenever it is possible. It can be used in an hosted or hostless approach

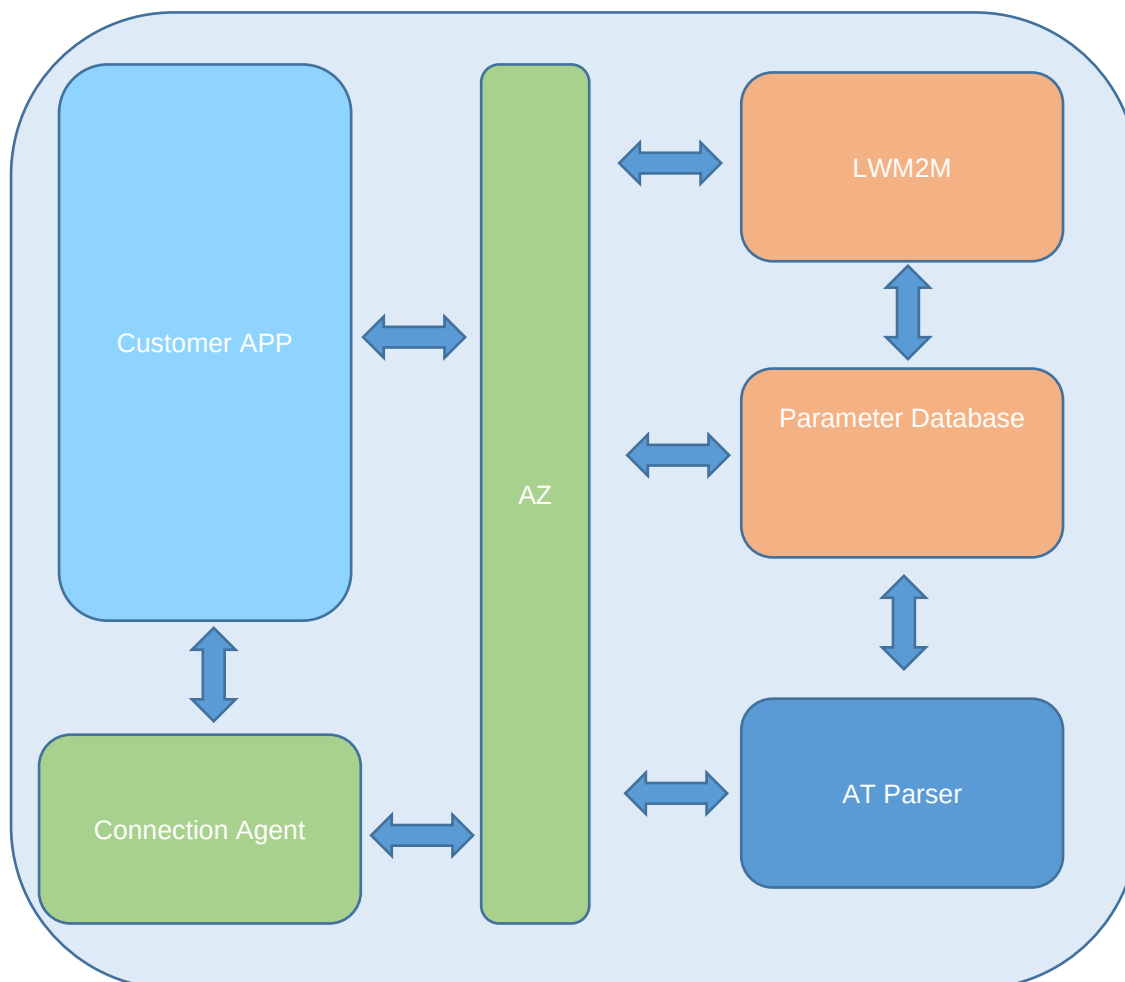
### 2.2. Hostless Approach

Below the module SW architecture of the Hosted version that show how the connection manager is linked to the main SW blocks



### 2.3. Hosted

In this architecture the connection manager will extend the set of AT commands in order to give to the host the possibility to automatically manage the connection.



In this architecture the connection manager will export a set of APIs dedicated to provide to the customer app the possibility to manage automatically the connection.

## 2.4. Limitations



As per version CA\_AT.00.xx1-6, the Connection Agent is running as one of the AppZone apps on the user space of the module. This means that by default is set with the run flag enabled (refer to AT#M2MRUN command). If a developer wants to run another app in parallel with the Connection Agent, AT#M2MRUN=2,<userapp.bin> **MUST NOT** be used to enable the user app, because this will stop the Connection Agent from running. User should use instead AT#M2MRUN=4,<userapp.bin>

### 3. MULTI SIM MAIN FLOW DIAGRAM

#### 3.1. Description

This is the description of the Dual SIM Management (abbreviated “DSM”) which is implemented in the Connection Agent.

The system provides a multi SIM management in the CoA to handle the connection to cellular network using the high level priority for one of the SIM inserted.

The system always try to use the preferred SIM (Default is 1 (one)).

If the DSM is enabled and signal is below threshold for 2 x antenna signal timeout, switch to backup sim card, change apn, possibly change technology, reset network for attach (e.g. nbloT), bootstrap for lwm2m if it was enabled.

If also backup sim is below the threshold, behave as in single mode on the backup sim for 5 times, then switch back to preferred card and disable dual SIM mode.

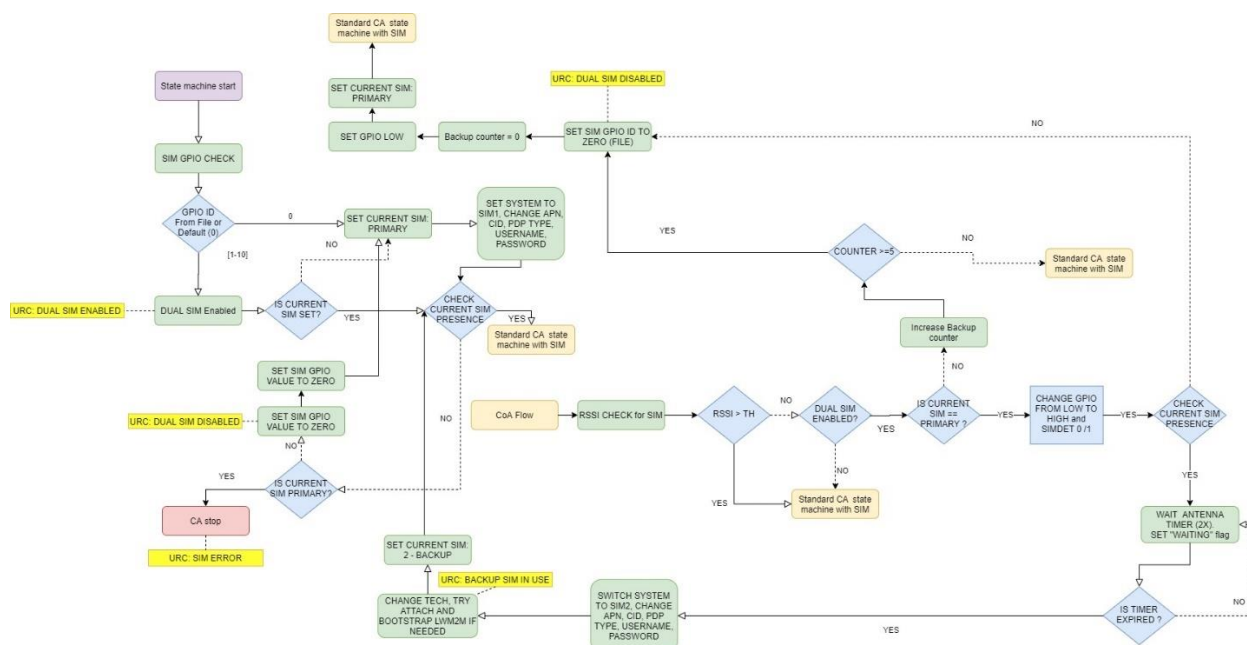
If the backup sim is not present, disable dual sim, switch back to preferred and send URC message “dual sim error” -> fallback to single SIM mode.

Is possible enable or disable the dual sim system handling a specific parameter of the CoA named: **#GPIO\_DSM**

This parameter allow to select a GPIO which will be used to switch throw the SIM cards.

#### 3.2. Block Diagram

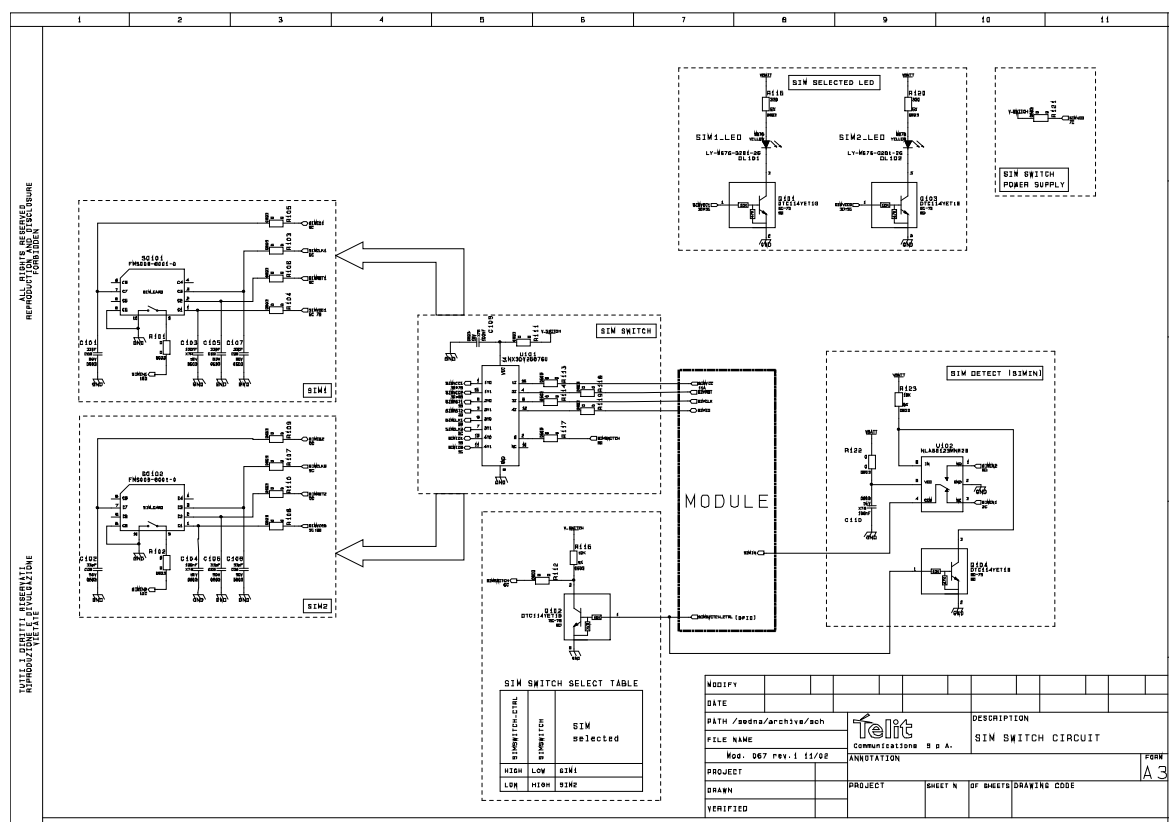
The diagram below shows the full flow of a proposed implementation for requirements described above.



## 4. HARDWARE DIAGRAM

### 4.1. Hardware Proposal

This is the basically the hardware proposal using two sim card readers and one sim switch:



### 4.2. GPIO Configuration

The selection between SIM cards will be possible through a GPIO on board.

It will be selected by the customer but the main logic will be:

SIMCARD1 = GPIO#XX LOW

SIMCARD2 = GPIO#XX HIGH

Default is set to LOW the GPIO, so the SIM1 is used by default.

Example Using AT Commands from SIM1 to SIM2 using GPIO2:

AT#SIMDET=0

AT#GPIO=2,1,1

AT#SIMDET=1

Example Using AT Commands from SIM2 to SIM1 using GPIO2:

```
AT#SIMDET=0
```

```
AT#GPIO=2,0,1
```

```
AT#SIMDET=1
```

The Connection Agent will use the same logic as AT Commands

## 5. AT COMMANDS REFERENCES

### 5.1. AT#CASN - Start and Stop the Connection Agent

This command allows to start and stop the execution of the agent, using previously set parameters.

| SIM Presence | Setting saved | Can be aborted | MAX timeout | SELINT |
|--------------|---------------|----------------|-------------|--------|
| Required     | -             | No             | -           | 2      |



**AT#CASN=<mode>**

Set command allows to starts/stops the Agent.

Parameter:

| Name | Type | Default | Description |
|------|------|---------|-------------|
|------|------|---------|-------------|

|        |         |   |                                  |
|--------|---------|---|----------------------------------|
| <mode> | integer | - | Connection Agent operation mode. |
|--------|---------|---|----------------------------------|

Value:

0 : Stop the Agent

1 : Start the Agent

Additional info:

- ▶▶ Depending on #URC configuration (see **AT#CACFG**), unsolicited messages will be shown on AT instances, notifying events occurrence.



**AT#CASN=?**

Test command reports the available range of values for parameter <mode> in the format:

**#CASN: (list of supported <mode> values)**



## 5.2. AT#CAVER - Check the version of the connection agent

This command allows verify the Connection Agent release version

| SIM Presence | Setting saved | Can be aborted | MAX timeout | SELINT |
|--------------|---------------|----------------|-------------|--------|
| Not Required | -             | No             | -           | 2      |



### AT#CAVER

If executed, this command return the current Connection Agent software version, as a string.

Additional info:

►► The version format returned will be in the format

**CA.MM.m**

**CA\_AT.NN.VVn[-BB]**

**OK**

Where

**MM** identifies the major release and **m** the minor release of the agent application library

**NN** identifies the major release, **VV** the module variant (e.g. 30 for ME910) and **n** the minor release of the Agent AT application. A optional **-BB** indicating the beta version can be present

## 5.3. AT#CACFG - Connection Agent parameters

This command allows manage Connection Agent parameters

| SIM Presence | Setting saved | Can be aborted | MAX timeout | SELINT |
|--------------|---------------|----------------|-------------|--------|
| Not Required | No            | No             | -           | 2      |



**AT#CACFG=<tag>,<v>**

Set command allows to configure each available parameter.

Parameter:

| Name  | Type   | Default     | Description                                                                                                                                                                                                                                                                                                                |
|-------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <tag> | String | N/A         | Connection Agent parameter tag.                                                                                                                                                                                                                                                                                            |
|       |        | Value:      |                                                                                                                                                                                                                                                                                                                            |
|       |        | <b>#LOG</b> | <p>Enable or disable the level of the debug logs on the trace port.</p> <p>Values admitted [0-4]</p> <p>0 – logs disabled</p> <p>1 – show up to errors</p> <p>2 – show up to info</p> <p>3 – show also debug messages</p> <p>4 – enable all messages</p> <p>Default: 0. Applies immediately.</p>                           |
|       |        | <b>#URC</b> | <p>URC messages verbosity,</p> <p>0 (No URC) to 2 (URC with max verbosity)</p> <p>Default 2.</p> <p>0 – No URC</p> <p>1 – URC in the format #CAEV: TYPE,code[,&lt;additional info&gt;]</p> <p>2- URC in the format</p> <p>#CAEV:</p> <p>TYPE_TAG,code,string_tag[,&lt;additional info&gt;]</p> <p>Applies immediately.</p> |

|                     |                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | For any information see chapter n.4                                                                                                                                                                      |
| <b>#APN</b>         | APN to be used for PDP context activation. Applies at next PDP activation.                                                                                                                               |
| <b>#APN_S2</b>      | APN to be used for PDP context activation for Backup SIM. Applies at next PDP activation.                                                                                                                |
| <b>#CID</b>         | Context ID to be used for PDP context activation, 1 to 6.<br>Default: Depends on Network operator:<br>Generic: 1<br>AT&T: 1<br>Verizon: 3<br>Applies at next PDP activation,                             |
| <b>#CID_S2</b>      | Context ID to be used for PDP context activation for Backup SIM, 1 to 6.<br>Default: 1<br>Applies at next PDP activation,                                                                                |
| <b>#PDP_TYPE</b>    | Indicates the type of PDP context to be activated<br>0 – IPv4 only<br>1 – IPv6 only<br>2 – IPv4 and IPv6 (depending on network operator)<br>Default: 2<br>Applies at next PDP activation.                |
| <b>#PDP_TYPE_S2</b> | Indicates the type of PDP context to be activated for Backup SIM<br>0 – IPv4 only<br>1 – IPv6 only<br>2 – IPv4 and IPv6 (depending on network operator)<br>Default: 2<br>Applies at next PDP activation. |

|                           |                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>#MAIN_TMR</b>          | <p>Main loop timer duration, in seconds. This is the period of the normal operation loop duration.</p> <p>Default: 10</p> <p>Applies immediately</p>                                                                                                                                     |
| <b>#SIGLEV_TMR</b>        | <p>Signal level check timer duration, in seconds. Must be greater than #MAIN_TMR. If the module does not have a signal level over the threshold an action can be taken at the expiration of this timer.</p> <p>Default: 20</p> <p>Applies immediately</p>                                |
| <b>#NETREG_TMR</b>        | <p>Network registration check timer duration, in seconds. Must be greater than #SIGLEV_TMR. If the module is not registered to a network an action can be taken at the expiration of this timer.</p> <p>Default: 50</p> <p>Applies immediately</p>                                       |
| <b>#DATA_SERV_TMR</b>     | <p>PDP data service check timer duration, in seconds. Must be greater than #NETREG_TMR. If the module does not have a valid PDP context an action can be taken at the expiration of this timer.</p> <p>Default: 60</p> <p>Applies immediately</p>                                        |
| <b>#SHDN_INTERVAL_TMR</b> | <p>Shutdown timer duration in seconds (will notify a critical condition that might require the module shutdown, for example missing PDP context, registration or valid signal level at expiration. Must be bigger than #DATA_SERV_TMR</p> <p>Default: 100</p> <p>Applies immediately</p> |
| <b>#RSSI_THRESH</b>       | <p>Signal level threshold for 2G fallback network, 0 to 31 (refer to AT+CSQ command). If the read</p>                                                                                                                                                                                    |

|                     |                                                                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>signal is below the threshold, the agent will consider the antenna level too low and avoid trying registration check and PDP activation.</p> <p>Default: 13</p> <p>Applies immediately</p>                                                                                                              |
| <b>#RSRP_THRESH</b> | <p>Signal level threshold for CAT-M1 and nbIoT networks, 0 to 97 (refer to AT+CESQ command). If the read signal is below the threshold, the agent will consider the antenna level too low and avoid trying registration check and PDP activation.</p> <p>Default: 30</p> <p>Applies immediately</p>        |
| <b>#BER_THRESH</b>  | <p>Bit Error rate threshold in 2G fallback network, 0 to 7 (refer to AT+CSQ command). If the read signal is below the threshold, the agent will consider the antenna level too low and avoid trying registration check and PDP activation.</p> <p>Default: 3</p> <p>Applies immediately</p>                |
| <b>#RSRQ_THRESH</b> | <p>Signal quality level threshold in CAT-M1 and nbIoT networks, 0 to 34 (refer to AT+CESQ command). If the read signal is below the threshold, the agent will consider the antenna level too low and avoid trying registration check and PDP activation.</p> <p>Default: 19</p> <p>Applies immediately</p> |
| <b>#AUTOSTART</b>   | <p>If 1, the connection agent will start at module boot.</p> <p>If 0, AT#CASN=1 must be executed.</p> <p>Default: 0</p>                                                                                                                                                                                    |

|                          |                                                                                                                                                                                                                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Applies at the next reboot of the agent                                                                                                                                                                                                                                                                     |
| <b>#USERNAME</b>         | <p>APN username.</p> <p>Default: ""</p> <p>Applies at the next PDP connections</p>                                                                                                                                                                                                                          |
| <b>#PASSWORD</b>         | <p>APN password.</p> <p>Default: ""</p> <p>Applies at the next PDP connections</p>                                                                                                                                                                                                                          |
| <b>#USERNAME_S2</b>      | <p>APN username for Backup SIM.</p> <p>Default: ""</p> <p>Applies at next PDP activation.</p>                                                                                                                                                                                                               |
| <b>#PASSWORD_S2</b>      | <p>APN password for Backup SIM.</p> <p>Default: ""</p> <p>Applies at next PDP activation.</p>                                                                                                                                                                                                               |
| <b>#POLICY</b>           | <p>Agent policy:</p> <ul style="list-style-type: none"> <li>0- PDP always on : the agent will try to keep PDP context always enabled</li> <li>1- Registration always on: the agent will try to keep up the registration.</li> <li>2- PSM/eDRX enable</li> </ul> <p>Default: 0</p> <p>Applies immediatly</p> |
| <b>#RETRY_MECH_PDP</b>   | For future use                                                                                                                                                                                                                                                                                              |
| <b>#ROAMING_ALLOW</b>    | <p>If 1, the agent allows registration in roaming, If 0, roaming will be denied.</p> <p>Default: 1</p> <p>Applies immediatly</p>                                                                                                                                                                            |
| <b>#ROAMING_ALLOW_S2</b> | <p>If 1, the agent allows registration in roaming, If 0, roaming will be denied (for Backup SIM).</p> <p>Default: 1</p> <p>Applies immediately</p>                                                                                                                                                          |

|                   |              |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | #NET_STATS   | <p>Enable or disable the collecting of agent statistics</p> <p>Default: 1</p>                                                                                                                                                                                                                                                                                                   |
|                   | #IOT_TECH    | <p>Select IoT technology to be used:</p> <ul style="list-style-type: none"> <li>0- CatM1</li> <li>1- nBloT</li> <li>2- Both</li> </ul> <p>Default: 0</p> <p>Only this setting is automatically stored and needs a reboot of the module* </p> <p>Applies at the next reboot of the module</p> |
|                   | #IOT_TECH_S2 | <p>Select IoT technology to be used for Backup SIM:</p> <ul style="list-style-type: none"> <li>3- CatM1</li> <li>4- nBloT</li> <li>5- Both</li> </ul> <p>Default: 2</p> <p>Applies at the next reboot of the module* </p>                                                                  |
|                   | #LWM2M_EN    | <p>If 1, the connection agent will notify its own status to LWM2M agent at every start and stop.</p> <p>If 0, disable the notifications.</p> <p>Default: 0</p> <p>Applies immediatly</p>                                                                                                                                                                                        |
|                   | #SIM_GPIO    | <p>Select which GPIO to be used to choose which SIM holder to be used.</p> <p>Values admitted [0-10]</p> <p>0 – Dual sim management disabled</p> <p>Default: 0.</p>                                                                                                                                                                                                             |
| <v> Number/string | N/A          | <p>Connection Agent parameter value,</p> <p>Value:</p> <p>Depending on TAG, see each one description.</p>                                                                                                                                                                                                                                                                       |

Additional info:



Parameters are NOT saved automatically. To store them in NVM, call **AT#CASAVE** command



Disable roaming  
**AT#CACFG=#ROAMING\_ALLOW,0**



**AT#CACFG?**

Read Command reports the current configuration of all available parameters, in the format

**<tag>: <v>**

**<tag>: <v>**

....

**OK**

For example:

**AT#CACFG?**

**#LOG: 4**

**#URC: 2**

**#APN: "apn.example.ext"**

**#APN\_S2: "apn2.example.ext"**

**#CID: 1**

**#CID\_S2: 1**

**#PDP\_TYPE: 2**

**#PDP\_TYPE\_S2: 2**

**#MAIN\_TMR: 5**

**#SIGLEV\_TMR: 20**

**#NETREG\_TMR: 50**

**#DATA\_SERV\_TMR: 60**

**#SHDN\_INTERVAL\_TMR: 100**

**#RSSI\_THRESH: 13**

**#RSRP\_THRESH: 30**

**#BER\_THRESH: 3**

**#RSRQ\_THRESH: 19**



```
#AUTOSTART: 0
#USERNAME: "username1example"
#USERNAME_S2: " username2example "
#PASSWORD: "password1example"
#PASSWORD_S2: " password2example "
#POLICY: 0
#RETRY_MECH_PDP: 0
#ROAMING_ALLOW: 1
#ROAMING_ALLOW_S2: 1
#NET_STATS: 1
#IOT_TECH: 0
#IOT_TECH_S2: 0
#LWM2M_EN: 0
#GPIO_DSM: 2
```



**AT#CACFG=?**

Test command reports the available range of values for parameter <tag> in the format:

```
#CACFG:
<tag>: (range)
<tag>: (range)
...
```

**OK**



**\*AT#CACFG=#IOT\_TECH,<value>**

This command automatically set and save in memory this value.

After this command a reboot of the module is necessary in order to change technology, otherwise the agent will continue to use the previous technology.

After this command and after the reboot, the module's technology will be changed permanently. For a future technology management will be possible resend this command, or AT#WS46=<value> as written in the ME910 manual.

To take effect this command need always a module reboot, even if is sent while the agent is stopped or running.

## 5.4. AT#CASAVE - Save Parameter of the Connection Agent

This command allows to store the current parameters configuration in a file that will be automatically loaded at next startup.

| SIM Presence | Setting saved | Can be aborted | MAX timeout | SELINT |
|--------------|---------------|----------------|-------------|--------|
| Required     | Yes           | No             | -           | 2      |



### AT#CASAVE

When executed, this command will save the parameters in nvram.

## 5.5. AT#CASTATS - Connection Agent Statistics

This command allows to retrieve and manage statistics information

| SIM Presence | Setting saved | Can be aborted | MAX timeout | SELINT |
|--------------|---------------|----------------|-------------|--------|
| Required     | YES           | No             | -           | 2      |



### AT#CASTATS=<s>

Set command allows to save statistics to file.

Parameter:

| Name | Type    | Default | Description                       |
|------|---------|---------|-----------------------------------|
| <s>  | integer | N/A     | Connection Agent statistics save. |

Value:

1 : Save statistics on a file.

The path is:

/settings/conn\_agent/stats.trz

---

Additional info: To save the file the command must to be sent every time is needed in order to reduce the number of writes in the file system.

- ▶▶ Depending on #URC configuration (see **AT#CACFG**), unsolicited messages will be showed on AT instances, notifying events occurrence.
- 



#### **AT#CASTATS?**

When executed, the command reports the current statistics values, in the format

**#CASTATS:**

**TOTAL\_ITERATIONS:<>**

**RSSI\_FAIL:<>**

**REG\_FAIL:<>**

**PDP\_SUCCESS:<>**

**PDP\_FAIL:<>**

**GENERIC\_FAIL:<>**

**OK**

---

## 6. URC MESSAGES

### 6.1. Preliminary info

When an event occurs, the Connection Agent will print an unsolicited message.  
Its format depends on the configuration of #CACFG=#URC value.

If URC is set to 0, no messages will be showed.

If URC is 1, messages will have the format

**#CAEV: TYPE,CODE[,<additional\_data>]**

Where

**TYPE** can be

- 0 for 3GPP for specific cellular network errors
- 1 for network registration events
- 2 for PDP context events
- 3 for antenna events
- 4 for system related events
- 5 for SIM Events
- 6 for dual SIM system

**CODE** is a numeric code, refer to the following paragraphs for the possible values

<additional\_data> could be the IP address received during PDP activation

e.g.

**#CAEV: 0,5**

If URC is 2, messages will have the format

**#CAEV: TYPE\_TAG,CODE,TAG[,<additional\_data>]**

**TYPE\_TAG** can be

- 3GPP for specific cellular network errors
- NWK for network registration events
- ANT for antenna events
- DAT for PDP context events
- SYS for system related events
- SIM for SIM related events

e.g. **#CAEV: 3GPP,5,IMEI\_NOT\_ACCEPTED**

## 6.2. 3GPP events

| 3GPP Network events (network errors) in URC mode 2 |
|----------------------------------------------------|
| #CAEV: 3GPP,2,IMSI_UNKNOWN_HLR                     |
| #CAEV: 3GPP,3,ILLEGAL_DEVICE                       |
| #CAEV: 3GPP,4,IMSI_UNKNOWN_VLR                     |
| #CAEV: 3GPP,5,IMEI_NOT_ACCEPTED                    |
| #CAEV: 3GPP,6,ILLEGAL_ME                           |
| #CAEV: 3GPP,7,GPRS_SERVICE_NOT_ALLOW               |
| #CAEV: 3GPP,8,GPRS_SERVICE_NON_GPRS_NOT_ALLOW      |
| #CAEV: 3GPP,9,DEVICE_IDENTITY                      |
| #CAEV: 3GPP,10,IMPLICITY_DETACHED                  |
| #CAEV: 3GPP,11,PLMN_NOT_ALLOWED                    |
| #CAEV: 3GPP,12,LOCATION_AREA_NOT_ALLOWED           |
| #CAEV: 3GPP,13,ROAMING_NOT_ALLOWED                 |
| #CAEV: 3GPP,14,GPRS_SERVICES_NOT_ALLOWED_IN_PLMN   |
| #CAEV: 3GPP,15,NO_SUITABLE_CALLS_IN_LOCATION_AREA  |
| #CAEV: 3GPP,16,MSC_NOT_REACHABLE                   |
| #CAEV: 3GPP,17,NETWORK_FAILURE                     |
| #CAEV: 3GPP,20,MAC_FAILURE                         |
| #CAEV: 3GPP,21,MAC_FAILURE                         |
| #CAEV: 3GPP,22,CONGESTION                          |

|                                                                      |
|----------------------------------------------------------------------|
| #CAEV: 3GPP,23,GSM_AUTHENTICATION_UNACCEPTABLE                       |
| #CAEV: 3GPP,24,MBMS_BEARER_CAPABILITIES_INSUFFICIENT_FOR_THE_SERVICE |
| #CAEV: 3GPP,25,LLC_OR_SNDP_FAILURE                                   |
| #CAEV: 3GPP,26,INSUFFICIENT_RESOURCES                                |
| #CAEV: 3GPP,27,MISSING_OR_UNKNOWN_APN                                |
| #CAEV: 3GPP,28,UNKNOWN_PDP_ADDRESS_OR_PDP_TYPE                       |
| #CAEV: 3GPP,29,USER_AUTHENTICATION_FAILED                            |
| #CAEV: 3GPP,30,ACTIVATION_REJECTED_BY_GGSN                           |
| #CAEV: 3GPP,31,ACTIVATION_REJECTED_UNSPECIFIED                       |
| #CAEV: 3GPP,32,SERVICE_OPTION_NOT_SUPPORTED                          |
| #CAEV: 3GPP,33,REQ_SERVICE_OPTION_NOT_SUBSCRIBED                     |
| #CAEV: 3GPP,34,SERV_OPTION_TEMPORARILY_OUT_OF_ORDER                  |
| #CAEV: 3GPP,35,NSAPI_ALREADY_USED                                    |
| #CAEV: 3GPP,36,REGULAR_DEACTIVATION                                  |
| #CAEV: 3GPP,37,QOS_NOT_ACCEPTED                                      |
| #CAEV: 3GPP,38,CALL_CANNOT_BE_IDENTIFIED                             |
| #CAEV: 3GPP,39,REACTIVATION_REQUIRED                                 |
| #CAEV: 3GPP,40,NO_PDP_CTXT_ACTIVATED                                 |
| #CAEV: 3GPP,41,SEMANTIC_ERROR_IN_TFT_OPERATION                       |
| #CAEV: 3GPP,42,SYNTACTICAL_ERROR_IN_TFT_OPERATION                    |
| #CAEV: 3GPP,43,UNKNOWN_PDP_CTXT                                      |
| #CAEV: 3GPP,44,SEM_ERR_IN_PKT_FILTER                                 |

|                                                                            |
|----------------------------------------------------------------------------|
| #CAEV: 3GPP,45,SYNT_ERR_IN_PKT_FILTER                                      |
| #CAEV: 3GPP,46,PDP_CNTXT_WITHOUT_TFT_ACTIVATED                             |
| #CAEV: 3GPP,47,MULTICAST_GROUP_MEMBERSHIP_TIMEOUT                          |
| #CAEV: 3GPP,48,RETRY_ON_NEW_CELL_BEGIN                                     |
| #CAEV: 3GPP,50,PDP_TYPE_IPV4_ONLY_ALLOWED                                  |
| #CAEV: 3GPP,51,PDP_TYPE_IPV6_ONLY_ALLOWED                                  |
| #CAEV: 3GPP,52,SINGLE_ADDRESS_BEARERS_ONLY_ALLOWED                         |
| #CAEV: 3GPP,63,RETRY_ON_NEW_CELL_END                                       |
| #CAEV: 3GPP,81,INVALID_TRANSACTION_IDENTIFIER                              |
| #CAEV: 3GPP,95,SEMANTICALLY_INCORRECT_MESSAGE                              |
| #CAEV: 3GPP,96,INVALID_MANDATORY_INFORMATION                               |
| #CAEV: 3GPP,97,MSG_TYPE_NON_EXISTENT_OR_NOT_IMPLEMENTED                    |
| #CAEV: 3GPP,98,MSG_TYPE_NOT_COMPATIBLE_WITH_PROTOCOL_STATE                 |
| #CAEV: 3GPP,99,IE_NON_EXISTENT_OR_NOT_IMPLEMENTED                          |
| #CAEV: 3GPP,100,CONDITIONAL_IE_ERROR                                       |
| #CAEV: 3GPP,101,MSG_NOT_COMPATIBLE_WITH_PROTOCOL_STATE                     |
| #CAEV: 3GPP,111,PROTOCOL_ERROR_UNSPECIFIED                                 |
| #CAEV: 3GPP,112,APN_RESTRICTION_VALUE_INCOMPATIBLE_WITH_ACTIVE_PDP_CONTEXT |

### 6.3. Antenna Signal

| RSSI events (signal level) in URC mode 2 | Description                         |
|------------------------------------------|-------------------------------------|
| #CAEV: ANT,3000,ANT_ERROR                | Signal level is below the threshold |
| #CAEV: ANT,3100,ANT_OK                   | Signal level is OK                  |

### 6.4. Network Registration

| Registration events in URC mode 2    | Description                              |
|--------------------------------------|------------------------------------------|
| #CAEV: NWK,1101,NET_STATUS_REG       | Module registered in home network        |
| #CAEV: NWK,1105,NET_STATUS_REG       | Module registered in roaming             |
| #CAEV: NWK,1000,NET_STATUS_FAIL      | Not registered, not searching            |
| #CAEV: NWK,1002,NET_STATUS_SEARCHING | Not registered, searching a new operator |
| #CAEV: NWK,1003,NET_STATUS_FAIL      | Registration denied                      |
| #CAEV: NWK,1004,NET_STATUS_FAIL      | Unknown                                  |
| #CAEV: NWK,1005,NET_STATUS_FAIL      | Registration in roaming not allowed      |

### 6.5. PDP context

| Data context events in URC mode 2                    | Description                               |
|------------------------------------------------------|-------------------------------------------|
| #CAEV: DAT, 2010,DATA_SERVICE_DEACTIVATED            | Deactivation of PDP context succeeded     |
| #CAEV: DAT, 2000,DATA_SERVICE_FAIL                   | Activation of PDP context failed          |
| #CAEV: DAT, 2100,DATA_SERVICE_OK                     | PDP context is active                     |
| #CAEV: DAT, 2110,DATA_SERVICE_OK,<IP>                | PDP context is active, IP address changed |
| #CAEV:<br>DAT,2099,NO_SOCKET_ASSOCIATED_TO_AGENT_CID | No socket associated to agent CID         |

### 6.6. System Codes

| System related events in URC mode 2   | Description                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| #CAEV: SYS,4100,SET_PARAM_FROM_NVM_OK | Loading of parameters from NVM succeeded                                                                                            |
| #CAEV: SYS,4010,CA_SHUTDOWN           | The agent reached a unrecoverable condition (no signal, no registration, no PDP context during all the SHDN_INTERVAL_TMR expiration |



## 6.7. SIM Codes

| System related events in URC mode 2  | Description                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------|
| #CAEV: SIM,5000,SIM_NOT_READY        | SIM is not inserted, not ready or there is an internal ERROR in the SIM management.   |
| #CAEV: SIM,5130,DUAL_SIM_2_IN_USE    | SIM 2 in use                                                                          |
| #CAEV: SIM,5020,DUAL_SIM_2_NOT_READY | SIM 2 is not inserted, not ready or there is an internal ERROR in the SIM management. |
| #CAEV: SIM,6100,DUAL_SIM_ENABLED     | Dual SIM system is now enabled                                                        |
| #CAEV: SIM,6200,DUAL_SIM_DISABLED    | Dual SIM system is now disabled                                                       |

## 7. CONFIGURABLE LWM2M PARAMETERS

All this parameters of the agent are fully managed via LWM2M request.

| Parameter               | Range         | Default value               | Read/Write |
|-------------------------|---------------|-----------------------------|------------|
| CID                     | 1 – 6         | Depends on Network Operator | Read Write |
| CID_2                   | 1 – 6         | Depends on Network Operator | Read Write |
| APN NAME                |               | Empty                       | Read Write |
| APN_2 NAME              |               | Empty                       | Read Write |
| PDP type                | 0 – 2         | 2                           | Read Write |
| PDP type_2              | 0 – 2         | 2                           | Read Write |
| Log Level               | 0 – 4         | 0                           | Read Write |
| URC level               | 0 – 2         | 2                           | Read Write |
| Main timer              | 4 – 60        | 10                          | Read Write |
| Signal Level Timer      | MT < X < NRT  | 20                          | Read Write |
| Net Registration Timer  | SLT < X < DST | 50                          | Read Write |
| Data Service Timer      | DST < X < ITS | 60                          | Read Write |
| Interval Timer Shutdown | X > DST       | 100                         | Read Write |
| RSSI Threshold          | 0 – 31        | 13                          | Read Write |
| RSRP Threshold          | 0 – 97        | 30                          | Read Write |
| BER Threshold           | 0 – 7         | 3                           | Read Write |

|                                              |        |    |            |
|----------------------------------------------|--------|----|------------|
| RSRQ Threshold                               | 0 – 34 | 19 | Read Write |
| Policy                                       | 0 – 2  | 0  | Read Write |
| PDP Retry Mechanism                          |        |    | Read Write |
| Roaming Allowed                              | 0 – 1  | 1  | Read Write |
| Roaming Allowed 2                            | 0 – 1  | 1  | Read Write |
| Net Statistics                               | 0 – 1  | 1  | Read Write |
| Autostart                                    | 0 – 1  | 0  | Read Write |
| Network IoT_technology                       | 0 – 2  | 0  | Read Write |
| Agent Release                                |        |    | Read       |
| Username (APN)                               |        |    | Read Write |
| Password (APN)                               |        |    | Read Write |
| Username 2 (APN 2)                           |        |    | Read Write |
| Password 2 (APN 2)                           |        |    | Read Write |
| LWM2M Enable                                 | 0 – 1  | 0  | Read Write |
| GPIO                                         | 0-10   | 0  | Read Write |
| Agent Statistics Total Attempts              |        |    | Read       |
| Agent Statistics RSSI Fail count             |        |    | Read       |
| Agent Statistics Net Registration fail count |        |    | Read       |
| Agent Statistics PDP Success count           |        |    | Read       |
| Agent Statistics PDP Fail count              |        |    | Read       |
| Agent Statistics Generic error count         |        |    | Read       |
| Agent restart                                |        |    | Exec       |

## 8. USE CASES

### 8.1. Preliminary – Load Application

#### Setup

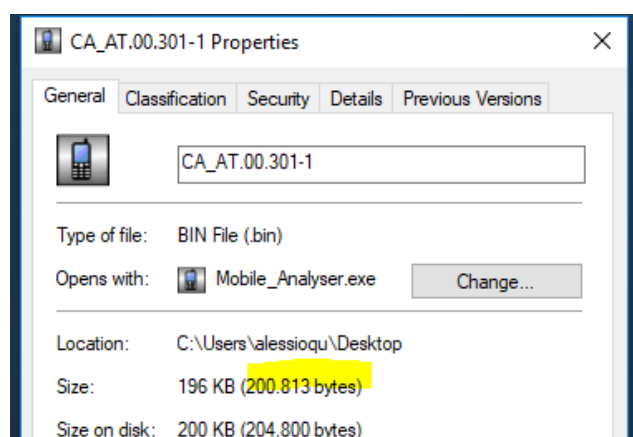
- 1- Connect to the device from an AT interface (USB or UART) using Telit AT Controller (TATC) and send

**AT#PORTCFG=0**

**AT#REBOOT**

- 2- Upload the Connection agent app on the module with

**AT#M2MWRITE=<file\_name.bin>,<size\_of\_file\_in bytes>,1**



*Figure 1 Application file size*

For example, given the file showed in Figure 1, send the following command in order to upload the app on the module

**AT#M2MWRITE=/mod/CA\_AT.00.301-1.bin,200813,1**

A prompt (>>>) will be showed, meaning the module is ready to accept file content.

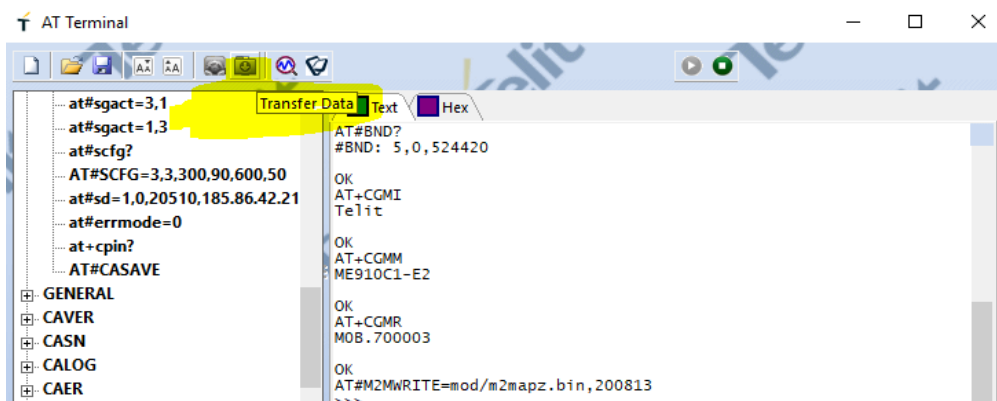


Figure 2 Transfer Data

Select from the top menu “Transfer Data” and browse to the folder where the application binary file is located and select ok.

The upload will start automatically.

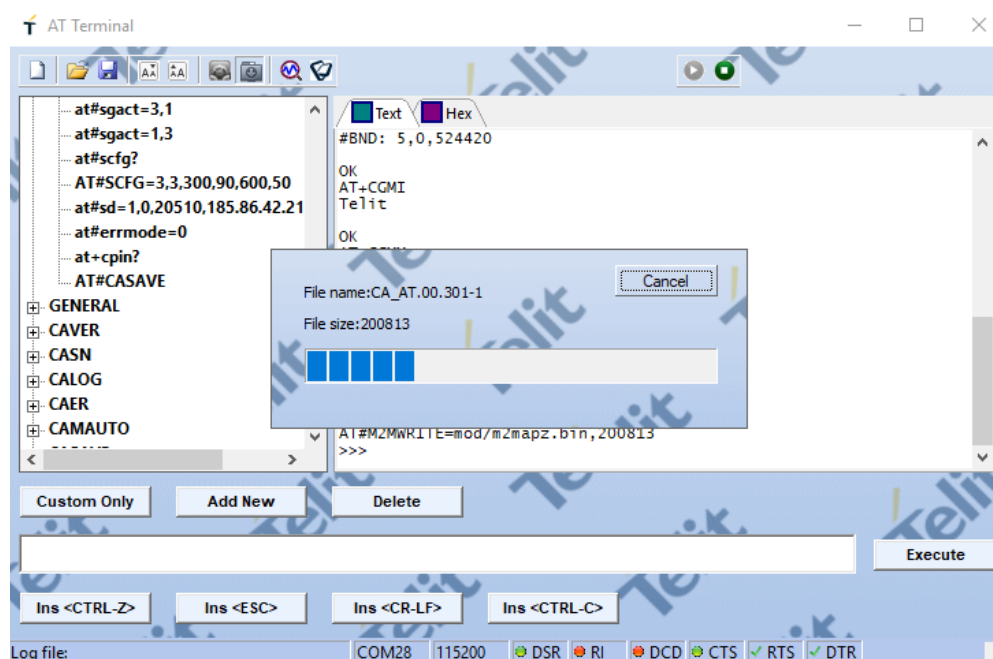


Figure 3 Data transferring

At the end of the process, a message with “OK” will be showed in the terminal



Figure 4 Upload complete

- 3- Enable the application to be executed with the command

**AT#M2MRUN=2,<file\_name.bin>**

In the example showed, the command would be

**AT#M2MRUN=2, CA\_AT.00.301-1.bin**

- 4- Reboot the device with:

**AT#REBOOT**

After the reboot, in order to see if the agent is started, check the version of the agent sending:

**AT#CAVER**

```
AT#CAVER
CA.00.1-11
CA_AT.00.301-11
OK
```

At this point, the connection agent application is ready for further configuration and management.

## 8.2. Use case #1 – PDP Always connected

### Scenario description:

PDP context must be always active, and ready for immediate socket operations right after module boot.

### Prerequisites

SIM card must be present, and unlocked (i.e. without PIN).

- 1- Configure the parameters for the specific scenario (must be done once)

**AT#CACFG=#APN,<custom apn>**

**AT#CACFG=#CID,<custom cid>**

**AT#CACFG=#URC,2**

**AT#CACFG=#AUTOSTART,1**

- 2- Save the Connection Agent parameters

**AT#CASAVE**

- 3- Restart the module

**AT#REBOOT**

After the reboot, a series of URC messages will be reported on the AT interface:

```
AT#CASN=1
OK
#CAEV: SYS,4100,SET_PARAM_FROM_NVM_OK
#CAEV: ANT,3100,ANT_OK
#CAEV: NWK,1101,NET_STATUS_REG
#CAEV: DAT,2100,DATA_SERVICE_OK
#CAEV: DAT,2110,DATA_SERVICE_OK,10.135.103.194
```

*Figure 5 URC messages in verbose format*

At this point, the agent is running and it is possible to use socket related AT commands, for example AT#SD.

Any other URC will be showed only in case of changes, for example

-Antenna signal error, in case of bad antenna signal level - It can be simulated by removing the antenna cable. The message below will be showed  
**#CAEV: ANT,3000,ANT\_ERROR**

-Network error in case of registration failure  
#CAEV: NWK,1004,NET\_STATUS\_FAIL

-Data error in case of context activation failure  
-IP address change

Every transition from a network state, e.g. a network registration, a bad antenna signal under a threshold value, deactivation of the pdp context, will be showed.

### 8.3. Use case #2 – Always registered to a network

#### *Scenario description:*

The module is already registered to a network but does not have a PDP context opened.  
The module is ready to establish a connection with the operator and to obtain a valid IP address.

#### *Prerequisites*

SIM card must be present, and unlocked (i.e. without PIN).

- 4- Configure the parameters for the specific scenario (must be done once)

**AT#CACFG=#APN,<custom apn>**

**AT#CACFG=#CID,<custom cid>**

**AT#CACFG=#URC,2**

**AT#CACFG=#POLICY,1**

- 5- (optional) Save the Connection Agent parameters

**AT#CASAVE**

- 6- Start the Connection Agent

**AT#CASN=1**

After the Agent startup, a series of URC messages will be reported on the AT interface (refer to previous use case).

At this point, the agent is running and it is possible to activate a context with AT commands, for example AT#SGACT=1,1. A valid ip address will be available.

After that the agent will be stopped sending

**AT#CASN=0**





## 9. ACRONYMS

| Item  | Description                                        |
|-------|----------------------------------------------------|
| APN   | Access Point Name                                  |
| BCCH  | Broadcast Control Channel                          |
| CSD   | Circuit Switched Data                              |
| CTM   | Cellular Text Telephone Modems                     |
| CTS   | Clear To Send                                      |
| DCE   | Data Circuit-Terminating Equipment (refer to [14]) |
| DRX   | Discontinuous Reception                            |
| DTE   | Data Terminal Equipment (refer to [14])            |
| DTMF  | Dual Tone Multiple Frequency                       |
| DTR   | Data Terminal Ready                                |
| GBR   | Guaranteed Bit Rate                                |
| GERAN | GSM EDGE Radio Access Network                      |
| GPIO  | General Purpose Input/Output                       |
| GUI   | Graphic User Interface                             |
| HF    | Hands Free (old terminology)                       |
| HS    | Hand Set (old terminology)                         |
| HSPA  | High Speed Packet Access                           |

|        |                                                          |
|--------|----------------------------------------------------------|
| IMS    | IP Multimedia Subsystem                                  |
| IRA    | International Reference Alphabet                         |
| ME     | Mobile Equipment                                         |
| MSISDN | Mobile Station International Subscriber Directory Number |
| NMEA   | National Marine Electronics Association                  |
| NVM    | Non-Volatile Memory                                      |
| PDN    | Public Data Network                                      |
| PDP    | Packet Data Protocol                                     |
| PDU    | Protocol Data Unit                                       |
| PIN    | Personal Identification Number                           |
| PPP    | Point to Point Protocol                                  |
| QoS    | Quality of Service                                       |
| SIM    | Subscriber Identification Module                         |
| SMS    | Short Message Service                                    |
| SMSC   | Short Message Service Center                             |
| TCP/IP | Transmission Control Protocol / Internet Protocol        |
| TTY    | Text Telephone Typewriter                                |
| UART   | Universal Asynchronous Receiver Transmitter              |
| UE     | User Equipment                                           |

|       |                                            |
|-------|--------------------------------------------|
| URC   | Unsolicited Result Code                    |
| USIM  | Universal Subscriber Identification Module |
| UTRAN | Universal Terrestrial Radio Access Network |

## 10. DOCUMENT HISTORY

| Revision | Date       | Changes                                                                 |
|----------|------------|-------------------------------------------------------------------------|
| 0        | 2019-02-12 | First emission                                                          |
| 1        | 2019-02-28 | Added limitations paragraph, updated firmware and app versions          |
| 2        | 2019-03-08 | Changed title of the use case n.2 and few lines in the same example     |
| 3        | 2019-03-27 | Changed some description errors in different points                     |
| 4        | 2019-03-29 | Changed Roaming description, applies immediately                        |
| 5        | 2019-04-05 | Changed AT instance value from 2 to 1                                   |
| 6        | 2019-04-09 | Changed Description of AT#CAVER command                                 |
| 7        | 2019-04-10 | Changed Main TMR value, RSSI_TH value and NET STATS value               |
| 8        | 2019-05-10 | Updated CACFG description and URC section.                              |
| 9        | 2019-06-27 | Changed SYSTEM_SHUTDOWN into CA_SHUTDOWN                                |
| 10       | 2019-07-04 | Added "applies information" in entire document for all agent parameters |
| 11       | 2019-07-11 | Added AT#CACFG=#IOT_TECH,<value> warning and description                |
| 12       | 2019-07-16 | Added new URC for socket cid                                            |
| 13       | 2019-08-07 | Changed #LWM2M_EN parameter                                             |
| 14       | 2019-08-27 | Inverted schema at page 10-11                                           |
| 15       | 2019-09-24 | Added path to CASTATS save file                                         |
| 16       | 2019-09-26 | Added lwm2m objects                                                     |
| 17       | 2019-09-30 | Document cleanup                                                        |
| 18       | 2019-10-10 | Changed description of policy 1, added SIM CA events                    |
| 19       | 2020-01-08 | Document cleanup                                                        |
| 20       | 2020-05-18 | Dual SIM                                                                |
| 21       | 2020-12-09 | Added supported modules                                                 |
| 22       | 2021-01-26 | Added a comment on STATS=1                                              |



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